

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2013-14

Paper No. GLB-503: Sedimentary and Metamorphic Petrology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/259

*Instructions: Attempt any **five** questions, selecting at least **two** questions each from Section A and Section B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Describe the process of diagenesis in sandstones. [14]

Question 2. With suitable sketches, describe the internal sedimentary structures. [14]

Question 3. Write notes on **any two** of the following: [2 × 7 = 14]

- (a) Ripple marks
- (b) Folk's classification of limestones
- (c) Sediment deposition

SECTION — B

Question 4. Describe the progressive metamorphism of Pelitic rocks in the $K_2O - FeO - MgO - Al_2O_3 - SiO_2$ system. [14]

Question 5. Discuss in detail the metamorphic facies. [14]

Question 6. Write notes on **any two** of the following: [2 × 7 = 14]

- (a) Retrograde Metamorphism
- (b) Phase rule
- (c) Principles of Metamorphic reactions

Question 7. Describe progressive metamorphism of Basic rocks in the $CaO - FeO - MgO - Al_2O_3 - SiO_2$ system. [14]

SECTION — C (Compulsory)

Question 8. Fill in the blanks or mark True (T) and False (F) as indicated: [7 × 2 = 14]

- (a) The typical product of contact metamorphism with maculose structure is _____.

- (b) Hot and humid climate promote physical weathering (T/F). _____
- (c) Charnockite is a typical rock of _____ facies.
- (d) Graywacke sandstones are mineralogically and texturally _____.
- (e) Foliation is defined by the _____ arrangement of minerals.
- (f) Flute casts are good palaeocurrent indicators (T/F). _____
- (g) Greenschist facies represents _____ grade metamorphism.